



BUSINESS REQUIREMENT SPECIFICATION

Timesheet data migration and business insights

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Executive Summary

The timesheet data migration ETL package is a SQL Server Integration Services (SSIS) solution that extracts consultant timesheet and leave data from excel spreadsheets. It transforms and stores the data into relational database tables in SQL server. It provides operational audit and monitoring logs and an automated pipeline to reduce manual data processing, improve data consistency and support business insights on their employees.

Project Objectives

- Automate the ingestion of consultant timesheet and leave data from excel spreadsheets.
- Process multiple consultant workbooks without requiring package redesign for each file.
- Validate source files and worksheet availability before processing.
- Maintain clean staging data for each pipeline execution.
- Synchronize staging data with production/master tables.
- Capture execution status, row-level processing metrics, and performance information.
- Provide operational dashboards for ETL health, SQL Server performance, and audit-log usage.
- Support controlled promotion from staging to production through reusable deployment automation
- Visualize timesheet insights in power BI report

Project Scope

This project aims to achieve the migration and transformation of consultant timesheet data from excel spreadsheets into a SQL server database using an SSIS ETL package. The system should be dynamic and easily configurable to be able to deploy this to a production or external environment like a co-worker's computer provided they have the relevant software installed (SQL server, Visual Studio and SQL Server data tools).

The system should also be monitored in the case of a pipeline failure, which is where the elastic stack comes in. SQL Server and SSIS analytics will be monitored here on dashboards where the user can track the SQL Server instance performance, each pipeline execution and data loading logs.

The ETL system is set on a SQL Server agent job schedule that will run the ISPAC file (the container holding the SSIS package within SQL Server) every 30 seconds ensuring data is always up to date. With the data located in SQL server, a Power BI analytics dashboard can be built around it to track and analyse the data like consultant performance or leave days taken

A Power BI report will be built around the newly transformed data to visualize and gain insights on the performance of consultants, leave and clients. This can make certain tasks for Operations simplified and can show the bigger picture in terms of understanding the data.

The Project does not account for error handling of dataset nor correction of consultant data. Payroll processing nor timesheet data entry. The system mainly focuses on the graduates'

timesheets as the scope is clearly defined but can be scaled with limitations to other consultants.

Project Constraints

The biggest constraint when building this project was time. Deadlines and schedules were very tight. Another issue was the lack of resources. Computing resources were low and our machines performed poorly while trying to keep up with the demanding workload. The latest version of Visual Studio also had bugs which led to the constant corruption of our projects leading us to restart. This was solved by the rollback of versions however it has set us back.

Functional Requirements

- Automatically identify and process Excel workbooks from the destination folder.
- Identify the consultant from workbook.
- Process only the correct timesheet worksheet.
- Check if the leave worksheet exists before processing.
- Exclude blank, invalid or irrelevant records.
- Load valid timesheet data into the timesheet staging table.
- Load valid leave data into the leave staging table.
- Clear staging tables before each processing cycle.
- Merge staged data into the master database.
- Record each ETL execution in the audit log.
- Automatically execute the SSIS package on a schedule.

Non-Functional Requirements

- Reliable: missing worksheets or invalid files should not unnecessarily stop processing.
- Performant: database indexes shall support lookups and merge operations.
- Maintainable: deployment logic shall be reusable and centralized.
- Secure: credentials shall be stored using GitHub Secrets.
- Auditable: ETL activity shall be traceable through Audit Log.
- Scalable: the solution shall support multiple consultants and workbooks.